Non-Gravitational Forces and Torques on LEO Satellites

Theory, Practical Considerations and Implementation

Reference document of the `NonGravForces` MATLAB toolbox, version 2.0.0 (concept 3)

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Abstract

This document collects the theory of the non-gravitational accelerations and torques acting on low-Earth-orbit satellites, the practical knowledge needed to model them at the $10^{-10}\,\mathrm{m/s^2}$ level for GRACE-FO accelerometer processing, and the implementation decisions of the `NonGravForces` toolbox: architecture, workflow, error handling, testing, performance, and the extension to future sensors (electrostatic, cold-atom interferometer, optomechanical, drag-free). It also records the audit of the two predecessor toolboxes and lists the test oracles every unit is verified against. Equations are given in the form the code implements; every approximation is named with its magnitude.

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1 Scope, notation and the force hierarchy

1.1 Scope

`NonGravForces` computes the non-gravitational acceleration $\boldsymbol{a}_{ng}$, the surface and body torques $\boldsymbol{\tau}$, and their partial derivatives, for a spacecraft described by a panel (macro) model, given its state, attitude and environment. It computes nothing else: time scales, Earth orientation, frame transformations, geodetic coordinates, ephemerides and attitude kinematics are the `referenceSystem` toolbox; Level-1B reading is `gracefoReading`; orbit integration is `gjToolbox`; parameter estimation is `lsqvce`. This partition (HANDOVER Part A.2) is the single most important design decision and it is what makes the toolbox smaller than either predecessor while covering more physics.

1.2 Notation

$\boldsymbol{r}, \boldsymbol{v}$ position and velocity in GCRS [m, m/s]; $\boldsymbol{q}$ attitude quaternion BODY→GCRS (scalar-first Hamilton, passive convention of `referenceSystem`); R_b the rotation matrix; m spacecraft mass; panel k with area A_k, outward unit normal $\hat{\boldsymbol{n}}_k$, centroid $\boldsymbol{c}_k$ (body frame), optical triplets (α, ρ_s, ρ_d) in the visible band and $(\varepsilon, \rho_s, \rho_d)$ in the infrared, energy accommodation α_E. $\hat{\boldsymbol{s}}$ unit vector to the Sun, $\hat{\boldsymbol{u}} = \boldsymbol{v}_r/|\boldsymbol{v}_r|$ the relative-flow direction, θ_k the incidence angle with $\gamma_k = \cos\theta_k = \hat{\boldsymbol{n}}_k \cdot (-\hat{\boldsymbol{u}})$ for drag and $\hat{\boldsymbol{n}}_k \cdot \hat{\boldsymbol{s}}$ for radiation. SI throughout; angles in radians; K epochs are stored as $3 \times K$ arrays.

1.3 Force hierarchy at the GRACE-FO altitude (∼490 km)

Effect	along-track	cross-track	source / note
atmospheric drag	5×10^{-8} (solar min) … 5×10^{-7} (max)	$\sim 10^{-9}$	Vielberg & Kusche 2020; Wöske et al. 2019
solar radiation pressure	13 nm/s^2 RMS	21 nm/s^2	Siemes et al. 2023, GRACE-A 2008
thermal re-radiation (TRP)	0.8 nm/s^2	6.4 nm/s^2	second-largest radiation term
Earth albedo	0.2 nm/s^2	0.2 nm/s^2	< 1 nm/s^2 combined with IR
Earth infrared	0.2 nm/s^2	0.1 nm/s^2	
thruster impulses	up to 10^{-6}		THR1B; Bandikova et al. 2019
CoM-offset coupling	$\dot{\boldsymbol{\omega}} \times \Delta\boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \Delta\boldsymbol{r})$		$\lvert\Delta\boldsymbol{r}\rvert \le 100$ µm (JPL D-56935)
antenna thrust, Lorentz	$\lesssim 10^{-11}$		hooks only
accelerometer noise floor	1×10^{-10} m s^{-2} Hz$^{-1/2}$	$\sim 10\times$ worse	Kornfeld et al. 2019

Table 1: Magnitudes in m/s^2 unless stated. Every unit of the toolbox is validated against these orders of magnitude before any finer oracle.

2 Atmospheric drag and lift

2.1 Free-molecular flow and the per-panel force

At 400–500 km the Knudsen number is $\gg 1$; molecules do not interact with each other near the surface and the force is the integral over the incident Maxwellian of the momentum exchanged at each panel. With mass density ρ, relative velocity $\boldsymbol{v}_r$ and per-panel pressure and shear coefficients $C_{p,k}, C_{\tau,k}$,

$$\boldsymbol{F}_{drag} = -\tfrac{1}{2}\rho\,|\boldsymbol{v}_r|^2 \sum_k A_k \left[C_{p,k}\,\hat{\boldsymbol{n}}_k + C_{\tau,k}\,\hat{\boldsymbol{t}}_k\right], \qquad \hat{\boldsymbol{t}}_k = \frac{\hat{\boldsymbol{u}} - (\hat{\boldsymbol{u}}\cdot\hat{\boldsymbol{n}}_k)\hat{\boldsymbol{n}}_k}{|\hat{\boldsymbol{u}} - (\hat{\boldsymbol{u}}\cdot\hat{\boldsymbol{n}}_k)\hat{\boldsymbol{n}}_k|}, \tag{1}$$

where the sign convention makes $C_p > 0$ push the panel inward along $-\hat{\boldsymbol{n}}_k$ for a windward panel. The projection onto $\hat{\boldsymbol{u}}$ is drag, the remainder is lift plus side force. The kernel returns the per-panel $\boldsymbol{F}_k$ (principle P2); $\boldsymbol{F}$ and $\boldsymbol{\tau} = \sum_k (\boldsymbol{c}_k - \boldsymbol{r}_{cm}) \times \boldsymbol{F}_k$ are reductions.

2.2 Relative velocity

$$\boldsymbol{v}_r = \boldsymbol{v} - \boldsymbol{\omega}_\oplus \times \boldsymbol{r} - \boldsymbol{v}_{wind}, \tag{2}$$

with the co-rotation term formed in `referenceSystem.applyRotation` (it carries $\boldsymbol{\omega} \times \boldsymbol{r}$ exactly, never by hand) and $\boldsymbol{v}_{wind}$ from HWM14 (Drob et al. 2015; quiet-time RMS 40–80 m/s) as an opt-in. At 7.6 km/s the co-rotation term is ∼470 m/s at the equator and enters cross-track drag at the 6 % level; winds of 100 m/s enter at 1.3 % of the along-track force but dominate the cross-track signal (Doornbos 2011).

2.3 Sentman's model (exact form implemented)

Let $s = |\boldsymbol{v}_r|/\sqrt{2\mathcal{R}T_\infty/M}$ be the speed ratio with M the mean molar mass of the local gas (*not* air: at 500 km atomic oxygen with $M \approx 16\,\mathrm{g/mol}$ dominates, so $s \approx 8$; using $R_{air} = 287\,\mathrm{J\,kg^{-1}\,K^{-1}}$ as v1 did biases s by $\sqrt{29/16} = 1.35$), $\gamma = \cos\theta$, $\ell = \sin\theta$, and

$$P = \frac{e^{-\gamma^2 s^2}}{s}, \quad Q = 1 + \frac{1}{2s^2}, \quad G = \frac{1}{2s^2}, \quad Z = 1 + \operatorname{erf}(\gamma s), \quad \frac{v_{re}}{v_{in}} = \sqrt{\tfrac{1}{2}\Big[1 + \alpha_E\Big(\frac{4\mathcal{R}T_w}{M|\boldsymbol{v}_r|^2} - 1\Big)\Big]}. \tag{3}$$

Then, after Sentman (1961) in the form of Doornbos (2011, Ch. 3),

$$C_p = \frac{P}{\sqrt{\pi}} + \gamma Q Z + \frac{\gamma}{2}\frac{v_{re}}{v_{in}}\big(\gamma\sqrt{\pi}Z + P\big), \tag{4}$$

$$C_\tau = \ell G Z + \frac{\ell}{2}\frac{v_{re}}{v_{in}}\big(\gamma\sqrt{\pi}Z + P\big). \tag{5}$$

The re-emission ratio v_{re}/v_{in} carries the wall temperature T_w and the accommodation coefficient α_E; complete accommodation ($\alpha_E = 1$) gives diffuse re-emission at T_w. Limits used as tests: $s \to \infty$, $\gamma = 1$ gives $C_p \to 2 + \ldots$ (flat plate); $\gamma \to 0$ gives the shear-only limit; $\gamma < 0$ (leeward) gives exponentially small terms. **Defect D1 of v1**: the exponential group of C_p and the whole shear term were divided by s instead of s^2, an error of $s \approx 8$ in the shear on inclined panels; the toolbox pins this as a regression test with the correct value from an independent kinetic-theory integration (`tools/pyxcheck/sentman_check.py`).

2.4 Other gas–surface interaction models

Maxwell: a fraction σ diffuse, $1 - \sigma$ specular. *Cercignani–Lampis–Lord (CLL)*: separate normal and tangential accommodation (α_n, σ_t) with quasi-specular lobes; used by Walker et al. (2014) and Mehta et al. (2014). *Schamberg*: quasi-specular lobe of adjustable width. All are `drag.gsi` options; none is in a preset except Sentman, because the density error (§2.6) exceeds their differences except below 300 km.

2.5 Energy accommodation

Published values: $\alpha_E \approx 0.85$ at 400–500 km (March et al. 2019, CHAMP/Swarm), 0.82 at 270 km (GOCE), 0.89 (CHAMP yaw, Pilinski & Argrow 2013). The SESAM model (Pilinski et al. 2010, 2013) makes α_E a function of the atomic-oxygen partial pressure p_O through a Langmuir isotherm $\Theta = Kp_O/(1 + Kp_O)$, $\alpha_E = \Theta\,\alpha_{ads} + (1 - \Theta)\,\alpha_{clean}$, reproducing measurements to ∼3 %. Options: `fixed` (default 0.85), `sesam`, `langmuir`.

2.6 Density and composition

Backends: exponential (sizing only), NRLMSISE-00 (Picone et al. 2002; the Brodowski C port, whose 17-case DOCUMENTATION table is the bit-exact oracle), NRLMSIS 2.0/2.1 (Emmert et al. 2021), DTM2020 operational/research (Bruinsma & Boniface 2021), JB2008 (Bowman et al.

2008). Inputs: $F_{10.7}$ (*observed*, previous day, and 81-day mean; v1 fed adjusted same-day values, ±3.4 %), F_{30}, $S_{10.7}, M_{10.7}, Y_{10.7}$ (JB2008, from SET), A_p 3-hourly history (the `ap7` vector), K_p/Hp, D_{st} for storms. Uncertainty: 10–30 % between models and data sets; NRLMSIS 2.1 is ∼10 % below NRLMSISE-00 at GRACE altitudes (Guo & Zhang 2024). Composition is required by the GSI (M, and p_O for SESAM). The models are evaluated at the *geodetic* height on WGS-84: a spherical Earth gives −21.4 km at the poles and a density factor of 1.40 on a polar mission (v1 defect D2); the conversion is `referenceSystem.cartesianToGeodetic` with the ellipsoid passed explicitly (its default is GRS80).

2.7 Macro model and self-shadowing

GRACE-FO (JPL D-56935 Table 5): eight plates, side panels at 50° with normals printed as $\sin 50° = 0.766044$, $\cos 50° = 0.642787$ (their squares sum to $1 - 5 \times 10^{-7}$; the toolbox accepts 10^{-5} and documents it), total area 16.916 328 4 m², mass 601.214 kg (C) / 601.209 kg (D), inertia tensor Table 4. The body is not convex: the flanks expose their undersides, so $\sum_k A_k \hat{\boldsymbol{n}}_k \neq 0$ and self-shadowing is physical. Wöske et al. (2019) quote ∼ 10 % force error of an 8-plate model against finite-element ray tracing. Self-shadowing options: none, icosphere look-up table over incidence directions (resolution a convergence-tested option), exact ray cast. The S-band boom is a cylinder (projection area 0.046 190 1 m², 0.27 %); it is a separate element kind, not a plate.

2.8 Implementation notes

Vectorise over K epochs: γ is a $N \times K$ array, erf and exp are evaluated once per array; the density backend accepts K points and returns ρ, species number densities and T_∞; the MSIS inner loop is the one place where hand-written C (MEX) is sanctioned for post-processing. NaN density is an error (`numericNaN`), never a zero force.

3 Solar radiation pressure

3.1 Irradiance and pressure

Total solar irradiance $S_0 = 1361\,\mathrm{W/m^2}$ at 1 au (IAU 2015 Resolution B3; SORCE/TIM; TSIS-1: 1361.5 ± 5.9), $P_0 = S_0/c = 4.5398 \times 10^{-6}\,\mathrm{Pa}$. The older $1367\,\mathrm{W/m^2}$ (Behzadpour et al. 2021, Table 1) is available as `Options.srp.tsi` for legacy comparisons and is never a constant (v1 used both values in two routines, 0.44 % inconsistency; v0.1.1 repeated the class of error between its oracle and its constant). Scale with $(\mathrm{au}/|\boldsymbol{r}_s - \boldsymbol{r}|)^2$; the Sun position comes from `referenceSystem.geocentricBodyState` (analytic SOFA tier, RMS 8.3×10^3 m; DE440 tier through MICE), never from a second implementation.

3.2 Per-panel optical model

For a panel with $\gamma = \hat{\boldsymbol{n}}_k \cdot \hat{\boldsymbol{s}} > 0$,

$$\boldsymbol{F}_k = -\nu\, P\, A_k\, \gamma \Big[(\alpha + \rho_d)\, \hat{\boldsymbol{s}} + \Big(2\rho_s \gamma + \tfrac{2}{3}\rho_d \Big) \hat{\boldsymbol{n}}_k \Big], \qquad \alpha + \rho_s + \rho_d = 1, \tag{6}$$

with $\nu \in [0, 1]$ the shadow function and 2/3 the Lambertian recoil factor. The same kernel serves Earth albedo (visible triplet) and Earth infrared (infrared triplet: $\varepsilon, \rho_s, \rho_d$) through the *source list* $\{\hat{\boldsymbol{d}}_j, P_j, \mathrm{band}_j\}$ (principle P3): SRP is one source, point-source Earth radiation two, a tiled Earth $2M$. Thermal re-emission of the absorbed part is *not* folded into eq. (6) by default (§5).

3.3 Shadow function

Apparent radii $a = \arcsin(R_s/|\boldsymbol{r}_s - \boldsymbol{r}|)$, $b = \arcsin(R_\oplus/|\boldsymbol{r}|)$, angular separation c between $-\hat{\boldsymbol{r}}$ and $\widehat{\boldsymbol{r}_s - \boldsymbol{r}}$. Cylindrical model: $\nu = 0$ iff $\boldsymbol{r} \cdot \hat{\boldsymbol{s}} < 0$ and $|\boldsymbol{r} - (\boldsymbol{r} \cdot \hat{\boldsymbol{s}})\hat{\boldsymbol{s}}| < R_\oplus$. Conical model (Montenbruck

& Gill 2000, §3.4.2): full light for $c \geq a + b$, umbra for $c \leq b - a$, annular ($\nu = 1 - b^2/a^2$) for $c \leq a - b$, else

$$x = \frac{c^2 + a^2 - b^2}{2c}, \quad y = \sqrt{a^2 - x^2}, \quad A = a^2 \arccos\frac{x}{a} + b^2 \arccos\frac{c - x}{b} - cy, \quad \nu = 1 - \frac{A}{\pi a^2}. \tag{7}$$

The conical function is continuous but only C^0 at the umbra edge; for orbit integration a C^1 transition is required or the integrator collapses its step. SOLAARS-CF (Robertson 2015; Zhang et al. 2019) adds Earth oblateness and atmospheric extinction/refraction as a tabulated attenuation curve; Siemes et al. (2023) quantify the model-version effect at 6–9 % of the radiation-pressure acceleration. Lunar eclipses use the same geometry with R_{moon} and are an opt-in. Tests: $\nu \in [0, 1]$, monotone across the penumbra, symmetric in the geometry, and the worked example of Montenbruck & Gill; a numerical disc integration in Python is the independent oracle.

4 Earth radiation pressure

4.1 Knocke–Ries–Tapley (1988)

Albedo and emissivity as zonal series with an annual term, $t_0 = 22$ December:

$$a(\phi, t) = a_0 + a_1 \cos\omega(t - t_0)\, P_1(\sin\phi) + a_2 P_2(\sin\phi), \quad e(\phi, t) = e_0 + e_1 \cos\omega(t - t_0)\, P_1(\sin\phi) + e_2 P_2(\sin\phi), \tag{8}$$

$a_0 = 0.34, a_1 = 0.10, a_2 = 0.29$; $e_0 = 0.68, e_1 = -0.07, e_2 = -0.18$; $\omega = 2\pi/365.25\,\mathrm{d}$. Each visible surface element dA at distance ϱ with outgoing angle ϑ is a Lambertian source: albedo flux $dP = \dfrac{a\, S_0 \cos\theta_s\, \cos\vartheta\, dA}{\pi c\, \varrho^2}$ (only for $\cos\theta_s > 0$, gated by the shadow function of the element), infrared flux $dP = \dfrac{e\, S_0 \cos\vartheta\, dA}{4\pi c\, \varrho^2}$. The visible cap is tiled (12×24 in `standard`, 24×48 in `precise`); a few hundred tiles suffice at LEO (convergence test).

4.2 CERES

Observed top-of-atmosphere fluxes (EBAF climatology, SYN1deg hourly) replace the zonal model at the $\mathrm{nm/s^2}$ level (Vielberg & Kusche 2020; Wang et al. 2023); anisotropic reflection through the CERES angular distribution models (Bhattarai et al. 2019) is the `erp.anisotropy=adm` option. The netCDF variable names must be read from a real file (open item O-3).

5 Thermal re-radiation

A panel at temperature T_k with emissivity ε_k emits $\varepsilon_k \sigma T_k^4 A_k$ and recoils with $\boldsymbol{F}_k = -\frac{2}{3}\,\varepsilon_k \sigma T_k^4 A_k\, \hat{\boldsymbol{n}}_k / c$. In steady state T_k follows from the balance $\alpha_k Q_{sun,k} + \alpha_k Q_{alb,k} + \varepsilon_k Q_{ir,k} + q_{int,k} = \varepsilon_k \sigma T_k^4$; the transient model adds $C_k \dot{T}_k$ with a per-panel heat capacity (Wöske 2019 thesis, Ch. 4; Rievers & Lämmerzahl 2011). For GRACE, TRP is the second-largest radiation term after SRP (Wöske et al. 2019; Vielberg & Kusche 2020; Siemes et al. 2023: $6.4\,\mathrm{nm/s^2}$ cross-track RMS), so "folding it into SRP" (the 2/3 factor on absorbed flux) is a documented, deliberately crude option, not the default. Metamorphic test: in steady state the total absorbed power equals the total emitted power.

6 Other effects and artefacts

Thruster forces: cold-gas attitude thrusters (THR1B) produce linear accelerations up to $10^{-6}\,\mathrm{m/s^2}$ that survive the 35 mHz CRN filter (Meyer et al. 2012; Bandikova et al. 2019 model them as impulse responses per thruster pair). The toolbox models them as impulses with the CRN response and uses THR1B epochs as *priors* for the impulse detector (§10). **Antenna thrust** (K-band, GPS): $\lesssim 10^{-11}\,\mathrm{m/s^2}$, hook only. **Lorentz force**: negligible at GRACE-FO, hook only. **Artefacts**

the model does not explain: twangs (up to $2 \times 10^{-5}\,\mathrm{m/s^2}$ radial; Flury et al. 2008; Peterseim 2014), heater switching, magnetorquer spikes (Peterseim et al. 2012). These are removed by quality control, never modelled.

7 Torques and the centre-of-mass coupling

Surface torque $\boldsymbol{\tau} = \sum_k (\boldsymbol{c}_k - \boldsymbol{r}_{cm}) \times \boldsymbol{F}_k$ from the same per-panel array (v1's separate torque implementation lost the vis/IR split: 8 %). Gravity gradient $\boldsymbol{\tau}_{gg} = \frac{3\mu}{r^3}\hat{\boldsymbol{r}} \times (\mathbf{I}\hat{\boldsymbol{r}})$ with $\hat{\boldsymbol{r}}$ in the body frame. Magnetic $\boldsymbol{\tau}_m = \boldsymbol{m} \times \boldsymbol{B}$ with residual and commanded dipoles (MAG1B) and $\boldsymbol{B}$ from IGRF-14 (IAGA 2024; degree 13 main field, degree 8 secular variation; oracle: the igrf14 test cases) evaluated in ITRS and rotated to the body frame.

The accelerometer proof mass sits at $\Delta\boldsymbol{r}$ from the centre of mass ($\leq 100\,\mu\mathrm{m}$ per axis by design, $\sim$50 µm achieved; GRACE-C CoM calibration agrees with JPL to 0.5 µm, Hu et al. 2022) and senses

$$\boldsymbol{a}_{sensed} = \boldsymbol{a}_{ng} + \dot{\boldsymbol{\omega}} \times \Delta\boldsymbol{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \Delta\boldsymbol{r}), \tag{9}$$

with $\boldsymbol{\omega}, \dot{\boldsymbol{\omega}}$ from SCA1B through `referenceSystem.angularVelocityFromRotationRate` after smoothing (Klinger & Mayer-Gürr 2016; Wöske 2019). This is a property of the *instrument*, not of the physics, and lives in the sensor layer (§8).

8 GRACE-FO data and the measurement model

8.1 Products and frames

ACC1A/ACT1A (10 Hz) → ACC1B/ACT1B (1 Hz) via time-tag correction and the CRN low-pass filter ($\sim$35 mHz, Wu et al. 2006); ACT (McCullough et al. 2019) substitutes thruster spikes and, for GRACE-D from 2018-06-21, transplants GRACE-C data (Bandikova et al. 2019); ACH (2022) hybridises. SCA1B quaternions are ICRF→SRF; the toolbox stores BODY→GCRS and conjugates exactly once in the attitude provider, verified by physics (nadir and line-of-sight), never by a flag. GNV1B is Earth-fixed (ITRS) and is rotated with `rotationItrsToGcrs` (v1 had no such path and treated GPS tags as UT1: 18 s of Earth rotation). MAS1B holds cold-gas tank masses ($\sim$30 kg), not spacecraft mass; the handbook constant is the default.

8.2 Calibration model

$\boldsymbol{a}_{cal} = \boldsymbol{b}(t) + \mathbf{S}\,\boldsymbol{a}_{meas}$ with a bias vector (cubic B-spline, 6-hour knots, Klinger & Mayer-Gürr 2016) and a full 3×3 scale/coupling matrix; typical scales 0.83–1.05 (GRACE-C along-track $1.05 \pm 2 \times 10^{-4}$, Siemes et al. 2023), biases $\sim 10^{-6}\,\mathrm{m/s^2}$ with thermal excursions of hundreds of $\mathrm{nm/s^2}$. Observability: along-track bias/scale well determined ($0.5\,\mathrm{nm/s^2}$, 1–3 %), cross-track poorly at solar minimum ($22\,\mathrm{nm/s^2}$, $\sim$15 %; van Helleputte 2009). The estimation is delegated to `lsqvce` (ARMA whitening from the sensor noise model, iterative snooping, variance components); `NonGravForces.app.calibrate` builds the observation vector and the design blocks, including Heaviside columns for detected jumps.

8.3 The sensor abstraction

The physics core returns the true dynamics record $\{\boldsymbol{a}_{ng}, \boldsymbol{\tau}, \boldsymbol{\omega}, \dot{\boldsymbol{\omega}}, t\}$; a `Sensor` maps it to an observable and provides the partials with respect to its own parameters and its one-sided noise ASD. **Electrostatic**: bias, scale matrix, CoM coupling, CRN filter; ASD $10^{-10}\sqrt{1 + (f/0.5)^4 + 0.005/f}\,\mathrm{m\,s^{-2}\,Hz^{-1/2}}$ (Kornfeld et al. 2019). **Atom interferometer** (CARIOQA pathfinder, MAGIC/NGGM hybrids): phase $\Delta\phi = k_{eff}\,a\,T^2$ with $k_{eff} = 4\pi/\lambda$, interrogation time T (1 s), cycle time (4 s, dead time), phase ambiguity modulo 2π requiring hybridisation with an electrostatic channel (Zahzam et al. 2022; Trimeche et al. 2019), and a rotation input because nadir pointing implies mrad/s

rates whose Coriolis phase must be compensated (Lévèque et al. 2022); projection-noise floor $10^{-10}\,\mathrm{m\,s^{-2}\,Hz^{-1/2}}$, target 10^{-12}. **Drag-free**: residual acceleration after control (LISA-Pathfinder heritage; NGGM $10^{-11}\,\mathrm{m\,s^{-2}\,Hz^{-1/2}}$ class). Optomechanical or MEMS instruments are further subclasses; calibration and quality control are written against the abstract interface, so no core code changes.

9 Partial derivatives

For orbit determination and calibration the toolbox returns $\partial\boldsymbol{a}/\partial\boldsymbol{r}$ (through $\partial\rho/\partial h$ and the shadow function), $\partial\boldsymbol{a}/\partial\boldsymbol{v}$ (drag, exact from eq. (1) since C_p, C_τ depend on $\boldsymbol{v}_r$ through s and γ), and parameter partials $\partial\boldsymbol{a}/\partial C_D = \boldsymbol{a}_{drag}/C_D$, $\partial\boldsymbol{a}/\partial C_R$, $\partial\boldsymbol{a}/\partial\alpha_E$. Analytic where clean, verified by complex-step differentiation $\mathrm{Im}\, f(x+ih)/h$ (no cancellation, 10^{-8} relative agreement is the test bound), finite differences only as a fallback. One evaluation serves both consumers (`Model.partials`).

10 Quality control: outliers, impulses, jumps

Three event classes with different signatures need three detectors, all operating on whitened residuals $r(t) = y_{meas} - (b + \mathbf{S}\, y_{model})$ from a preliminary calibration:

- **Outliers** (isolated samples): Hampel filter, moving median and 1.4826·MAD over a 61 s window, threshold 4σ; robust to 50 % contamination, so thruster spikes and twangs cannot inflate the scale (a moving mean/std would). Implemented and verified against an independent numpy oracle (`qc.hampel`).
- **Impulses** (CRN-shaped pulses, ∼30 s wide, simultaneous in three axes): matched filter with the CRN impulse response; THR1B epochs are assigned first, unassigned pulses are twang/impact candidates; an impact (not a twang) also changes $\boldsymbol{\omega}$ in SCA1B.
- **Jumps** (steps in level or slope): PELT change-point detection (Killick et al. 2012) with BIC penalty on piecewise-constant and piecewise-linear models; eclipse entries/exits (from the shadow function) and mass-trim events (MAS1B) are priors. Jumps are *estimated* as Heaviside columns in the calibration, never cut.

Output: a canonical event table (`qc.eventTable`: t_{start}, t_{end}, class, axis, amplitude, σ, source, confidence) and a logical mask. Limits: impulses shorter than the CRN width are unresolvable in L1B (use ACC1A); jumps below $\sim 1\,\mathrm{nm/s^2}$ are indistinguishable from $1/f$ bias drift without temperature data; along-track thresholds must adapt to geomagnetic activity because the residual carries the 10–30 % density error. Metamorphic tests: time shift, affine scaling, time reversal, planted events recovered with amplitude to 3σ.

11 Implementation

11.1 Architecture

Nine layers (L0 core, L1 io, L2 env, L3 geom, L4 kernel, L5 `Model`, L6 sensor, L7 qc, L8 app), 51 units, one catalogue (`doc/functionlist.csv`) that generates the function reference and drives the stub and audit tests. A layer calls only lower layers plus the four dependencies. `Model` is the single point of access (`accel | torque | series | partials | rhs | budget`); `Options` is a value class with presets `fast/standard/precise` and validated dotted overrides; extended models are opt-ins that no preset enables, so preset results never move when a model lands.

11.2 Workflow

```
run('.../referenceSystem/setupReferenceSystem.m'); run('.../gracefoReading/
    setupGraceFOReading.m')
setupNonGravForces; runNgfTests
```

```
[ep,r,v]  = NonGravForces.io.readGNV1B(gnv);      % ITRS -> GCRS via referenceSystem
[~,q]     = NonGravForces.io.readSCA1B(sca);      % ICRF->SRF conjugated once
[~,y]     = NonGravForces.io.readACC1B(act);
macro = NonGravForces.geom.graceFO("C");
opt   = NonGravForces.Options("standard","drag.wind","hwm14");
M     = NonGravForces.Model(macro, opt, 'Attitude', att, 'Mass', mass);
T     = M.series(ep, r, v, q);                    % a_ng, tau, provenance
sens  = NonGravForces.sensor.Electrostatic(...);
[mask, events] = NonGravForces.qc.screen(y, sens.observe(T), thr, eclipse, mass);
R = NonGravForces.app.calibrate(y, T, sens, 'Events', events, 'Mask', mask);  % lsqvce
```

11.3 Error handling

Every error passes through `NonGravForces.fail(unit, mnemonic, message)`; the identifier is `NonGravForces:<unit>:<mnemonic>`, the mnemonic's first token is one of `input, dep, data, numeric, unsupported, notImplemented`, and the message ends with a correction hint after " `->` "; `fail` refuses anything else. Contracts are `arguments` blocks (shape, class, range) plus semantic checks at the top of the body; nothing is validated twice (the macro model validates once, kernels trust its packed arrays). No fidelity fallback exists; unknown option fields error; absent optional dependencies error at the point of use naming their setup function; dependency errors propagate unwrapped. Every result carries `info.provenance` (toolbox and dependency versions, option hash, files, warnings, UTC time). NaN policy is stated per unit: a NaN density is an error because it would silently zero the drag; a NaN sample in quality control is never flagged and never poisons a neighbour. Numerical guards report their diagnostic (condition number, residual, epoch index). Tests assert identifiers, not messages, and pair every refusal with the case that must succeed.

11.4 Testing

Seven categories per public unit, counted mechanically: contract, reference (an authority outside the project: published tables, independent Python, real data), precision (a bound with the reason for its size in a comment), vectorization (K at once equals one at a time, and chunked/parallel series are bit-identical), robustness (NaN, gaps, empty, flags), metamorphic (rotation covariance, affine/shift/reversal invariance, mass and density scaling, $\nu \in [0, 1]$ monotone), speed (ratios against a built-in, absolute times recorded). A passing bound leaves its number in `tests/last_run.txt`; the gate is failures + incomplete + audit shortfall + written log, not the pass count. This session: 74 tests, GREEN in R2026a Update 4; two failures were authoring errors found by decomposition (a typed oracle value; a 0/1-based index), one was a guessed bound corrected by measurement (interpreted loop 382× slower than `movmedian`; algorithmically equivalent).

11.5 Performance

$3 \times K$ column layout; environment evaluated in bulk; `pagemtimes` for attitude; the panel loop over $N \approx 8$ with K-vectorised arithmetic; self-shadow LUT instead of per-epoch ray casting; `series` chunks K and optionally runs the chunks under `parfor` (Parallel Computing Toolbox, feature-detected) with bit-identical results either way; the single-epoch C/MEX kernel exists only for the integrator right-hand side, where MATLAB call overhead (5–15 µs) dominates, and links `referenceSystem`'s `rsApiRotation`; no GPU (eight panels give a GPU nothing to do).

12 Audit of the predecessors (summary; full table in HANDOVER Part C)

v1 (`SatelliteNonGravToolbox`): Sentman shear ×8; air gas constant; spherical Earth (−21.4 km, density ×1.40); two solar constants; GPS as UT1, no ITRS→GCRS; adjusted same-day $F_{10.7}$;

silent density fallback; drag routine computes its own Sun; five optical implementations, torque without vis/IR split; single-epoch interfaces; hand-rolled tests; duplicated frame helpers. **v0.1.1** (`NonGravToolbox`): the physics specification is correct; the flaws are architectural: gravity and equations of motion inside a non-gravitational toolbox, a compatibility layer to a defective v1, a Sun "selector" that is a wrapper, a 4171-line handover mixing specification and diaries, a setup that does not add the tests, four instead of seven test categories, a 1367 oracle against a 1361 constant, no sensor layer, no quality control, error handling by convention only. Concept 3 keeps v0.1.1's dependency discipline, principles, help template and ○/☒ rule, removes the scope creep, and adds the sensor and QC layers.

13 Recommended defaults and test oracles

Quantity	Default	Authority
density	NRLMSISE-00 (`standard`), NRLMSIS 2.1 (`precise`)	Picone 2002; Emmert 2021
wind	co-rotation; HWM14 opt-in	Drob 2015
GSI	Sentman, $\alpha_E = 0.85$; SESAM in `precise`	Sentman 1961; March 2019; Pilinski 2013
S_0	1361 W/m^2; 1367 override	IAU 2015 B3; SORCE/TIM
Lambert factor	2/3	standard
shadow	conical; SOLAARS-CF in `precise`	Montenbruck & Gill 2000; Robertson 2015
Earth radiation	Knocke 12×24; CERES 24×48	Knocke 1988; Wielicki 1996
TRP	steady-state per panel; transient opt-in	Wöske 2019
magnetic field	IGRF-14	IAGA 2024
ephemeris	SOFA analytic; DE440 in `precise`	`referenceSystem`
ellipsoid	WGS-84, passed explicitly	
GRACE-FO mass, geometry	601.214 kg; JPL D-56935 Tables 4–5	Wen 2019

Oracle	Used for
CODATA 2018, IAU 2015 B3	`const` (verified)
JPL D-56935 Tables 4–5	`geom.graceFO` (verified: areas, optics, inertia, normals)
numpy Hampel (`tools/pyxcheck`)	`qc.hampel` (verified, bit-for-bit flags)
Brodowski NRLMSISE-00 DOCUMENTATION table (17 cases)	density backend
NRL/CCMC NRLMSIS 2.x and HWM14 test outputs; CNES DTM2020; SET JB2008	extended density and wind
igrf14 test cases; Zenodo 10.5281/zenodo.14012303	magnetic field
Montenbruck & Gill §3.4.2 example; Python disc integration	shadow function
Knocke 1988 Table 1; Orekit `KnockeRediffusedForceModel`	Earth radiation
Python kinetic-theory integration; v1 ×8 regression	Sentman coefficients
Pilinski 2013 Fig. 3	SESAM
Siemes 2023 Table 3; Vielberg & Kusche 2020 monthly means	radiation magnitudes per axis
Hu 2022 CoM calibration; Kornfeld 2019 ASD	electrostatic sensor
Lévèque 2022 CARIOQA parameters	atom-interferometer sensor
real ACT1B/GNV1B/SCA1B/THR1B/MAS1B 2020-01-01 GRACE-C	readers, end-to-end calibration (scale ≈ 1.05)

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